Large Language Models for the
Economic and Social Sciences:

Content Analysis with LLMs

Indira Sen, HWS 2026
https://github.com/dess-mannheim/LLM4ESS-HWS25

Summary of Last Week and Today

Last session we learned:

- LLMs to LLM Agents
- Components of LLM Agents
- Evaluation of LLM Agents

Today:

- What is content analysis
- What do LLMs have to with it?
- Can LLMs really help with content analysis?

What is content analysis?

Why?

- Content analysis:
  - “any technique for making inferences by objectively and systematically identifying specified characteristics of messages” [Holsti, 1969]
  - “a systematic, replicable technique for compressing many words of text into fewer content categories based on explicit rules of coding” [Berelson, 1952; GAO, 1996; Krippendorff, 1980; Weber, 1990]

Why?

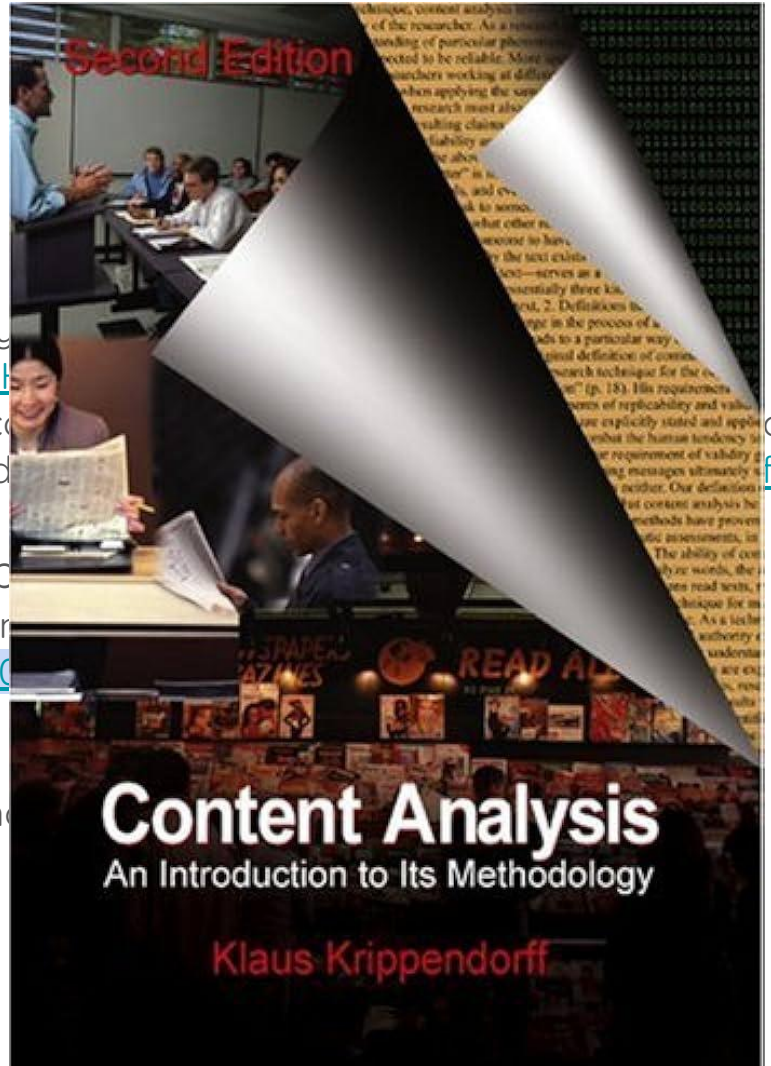
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- Questions content analysis can help us answer:
  - How does news media represent the immigration crisis?
  - What are topics that lead to arguments in long-term relationships?
  - How do citizens perceive the performance of politicians during the pandemic?

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- Questions content analysis can help us answer:
  - How does news media represent the immigration crisis?
    NYtimes articles frames [Mendelsohn, 2021]
  - What are topics that lead to arguments in long-term relationships?
    r/relationship_advice posts topics
  - How do citizens perceive the performance of politicians during the pandemic?
    Twitter posts stance

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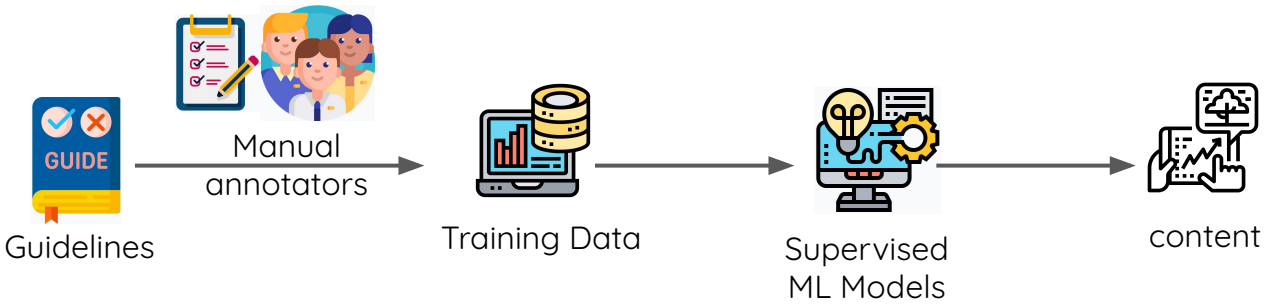


Before LLMs: ‘Modern Pipeline’

- unsupervised



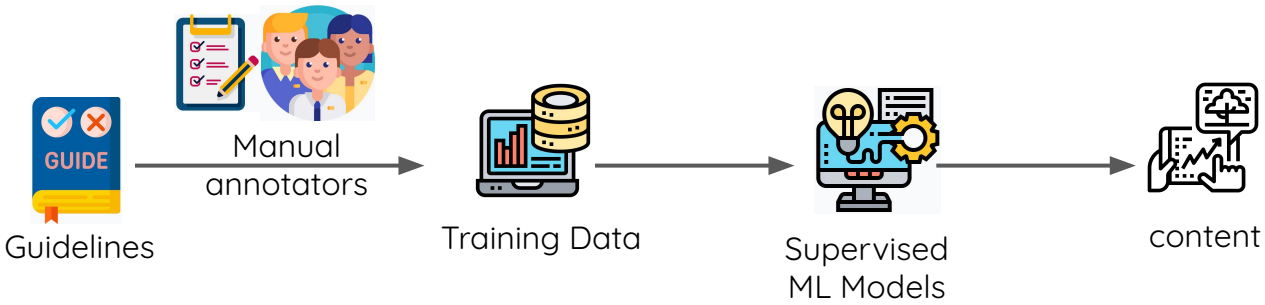
- supervised



Before LLMs: ‘Modern Pipeline’

- unsupervised

- supervised



Before LLMs: ‘Modern Pipeline’

- unsupervised

Unsupervised ML Models

Qualitative assessment

content

- supervised



Guidelines



Manual annotators

crowdworkers

amazon mechanical turk

Prolific

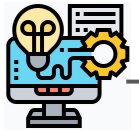
OR

Research assistants





Training Data



Supervised ML Models



content

Before LLMs: ‘Modern Pipeline’

- unsupervised

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Qualitative assessment

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Guidelines



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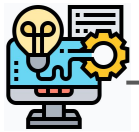
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Supervised ML Models



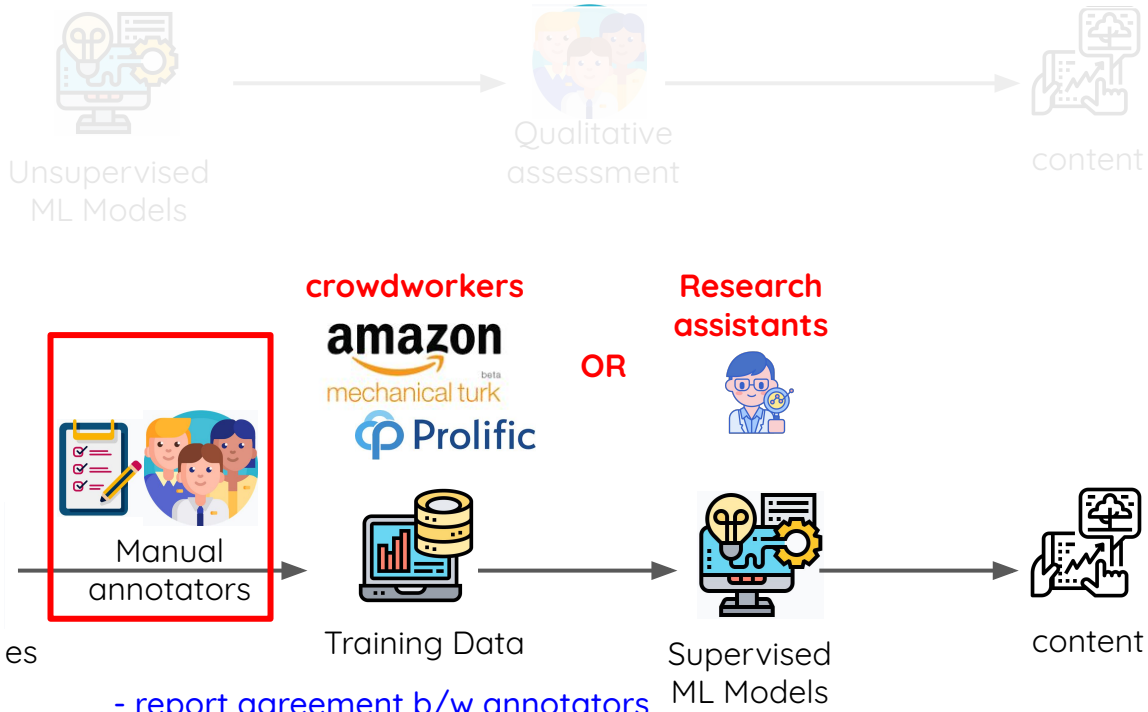
content

- report agreement b/w annotators
- throw out instance with disagreement

Before LLMs: ‘Modern Pipeline’

Not just useful for research, these content analysis models also applied for:

- **Ranking** content on platforms (what shows up on your feed)
- **Recommender systems**
- **Content moderation**
- Other types of **algorithmic decision making**



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Fast, cheap, and creative: Evaluating translation quality using Amazon's Mechanical Turk

Fast, Cheap, and Creative: Evaluating Translation Quality Using Amazon’s Mechanical Turk

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Who are the Crowdworkers? Shifting Demographics in Mechanical Turk

Who are the crowdworkers? Shifting demographics in Mechanical Turk

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M. Six Silberman
Department of Informatics

Abstract
Amazon Mechanical Turk (MTurk) is a crowdsourcing system in which tasks are distributed to a population of thousands of anonymous workers for completion. This system is increasingly popular with researchers and developers. Here we extend previous studies of the demographics and usage behaviors of MTurk workers. We describe how the worker population has changed over time, shifting from a primarily moderate-income, U.S.-based workforce towards an increasingly international group with a significant population of young, well-educated Indian workers. This change in population points to how workers may treat Turking as a full-time job, which they rely on to make ends meet.

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Sometimes, we have no information about crowdworkers. Better with RAs but not often reported

Fast, cheap, and creative: Evaluating translation quality using Amazon's Mechanical Turk

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A few other aspects to keep in mind

https://en.wikipedia.org/wiki/Inter-rater_reliability

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Inter-rater reliability

10 lang

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From Wikipedia, the free encyclopedia

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In statistics, **inter-rater reliability** (also called by various similar names, such as **inter-rater agreement**, **inter-rater concordance**, **inter-observe reliability**, **inter-coder reliability**, and so on) is the degree of agreement among independent observers who rate, code, or assess the same phe

Assessment tools that rely on ratings must exhibit good inter-rater reliability, otherwise they are not valid tests.

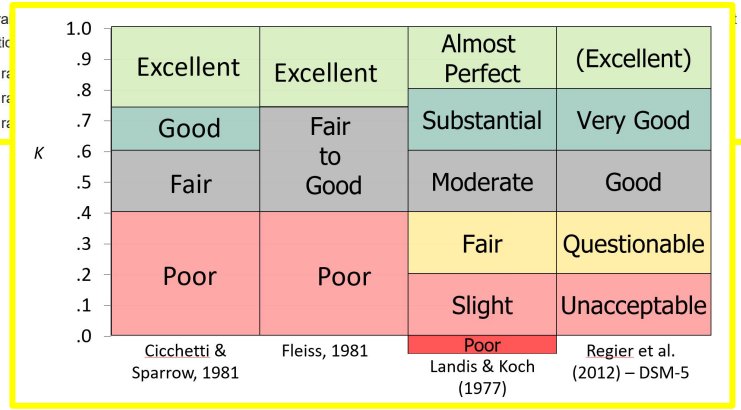
There are a number of statistics that can be used to determine inter-rater reliability. Different statistics are appropriate for different types of measu

Some options are joint-probability of agreement, such as Cohen's kappa, Scott's pi and Fleiss' kappa; or inter-rater correlation, concordance corre coefficient, intra-class correlation, and Krippendorff's alpha.

Concept [edit]

There are severa : between raters are three operatic

1. Reliable ra
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Annotator Agreement often used as proxy of 'feasibility of task'

Low agreement => raters cannot come to a consensus, so how could a model?

Inter-rater reliability

10 lang

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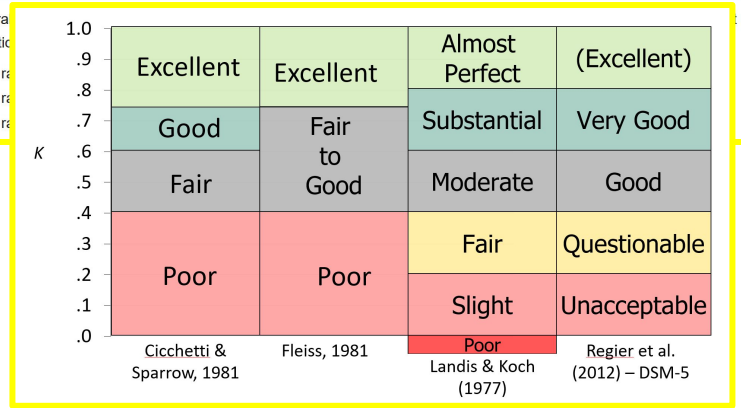
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Sentiment Analysis: It's complicated!

Sentiment Analysis: It's Complicated!

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Are We Modeling the Task or the Annotator? An Investigation of Annotator Bias in Natural Language Understanding Datasets

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Yoav Goldberg
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In this paper, we continue recent efforts to understand biases that are introduced during the pro... et al., 2015; Schwartz et al., 2018; Glockner et al., 2018; Poliak et al., 2018; Tsuchiya, 2018;

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Even for 'basic' tasks, annotators disagree

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The Risk of Racial Bias in Hate Speech Detection

Maarten Sap[◇] **Dallas Card**[♣] **Saadia Gabriel**[◇] **Yejin Choi**[◇♡] **Noah A. Smith**[◇♡]
[◇]Paul G. Allen School of Computer Science & Engineering, University of Washington, Seattle, USA
[♣]Machine Learning Department, Carnegie Mellon University, Pittsburgh, USA
[♡]Allen Institute for Artificial Intelligence, Seattle, USA
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Abstract

We investigate how annotators’ insensitivity to differences in dialect can lead to racial bias in automatic hate speech detection models, potentially amplifying harm against minority populations. We first uncover unexpected correlations between surface markers of African American English (AAE) and ratings of toxicity in several widely-used hate speech datasets. Then, we show that models trained on these corpora acquire and propagate these biases, such that AAE tweets and tweets by self-identified African Americans are up to two times more likely to be labelled as offensive compared to others. Finally, we propose dialect and race priming as ways to reduce the racial bias in annotation, showing that when annotators are made explicitly aware of an AAE tweet’s dialect they are significantly less likely to label the tweet as offensive.

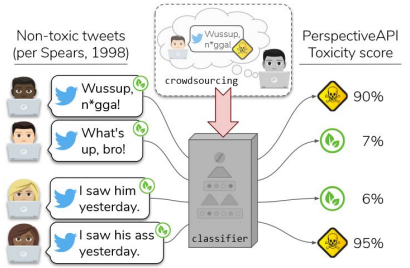


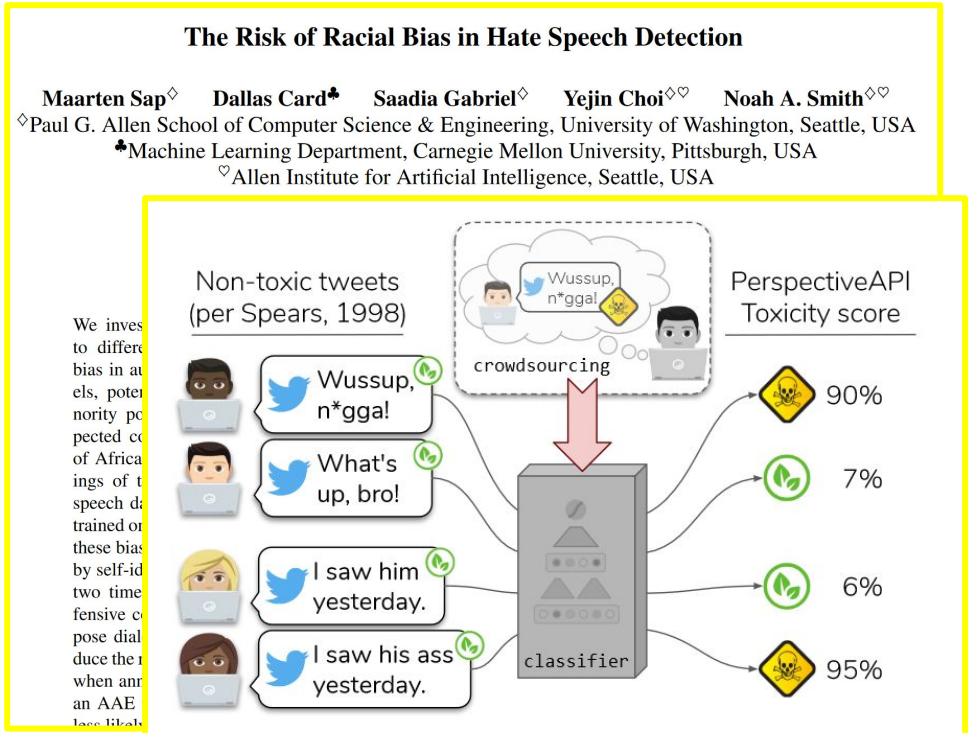
Figure 1: Phrases in African American English (AAE), their non-AAE equivalents (from Spears, 1998), and toxicity scores from PerspectiveAPI.com. Perspective is a tool from Jigsaw/Alphabet that uses a convolutional neural network to detect toxic language, trained on crowdsourced data where annotators were asked to label the toxicity of text without metadata.

The Risk of Racial Bias in Hate Speech Detection

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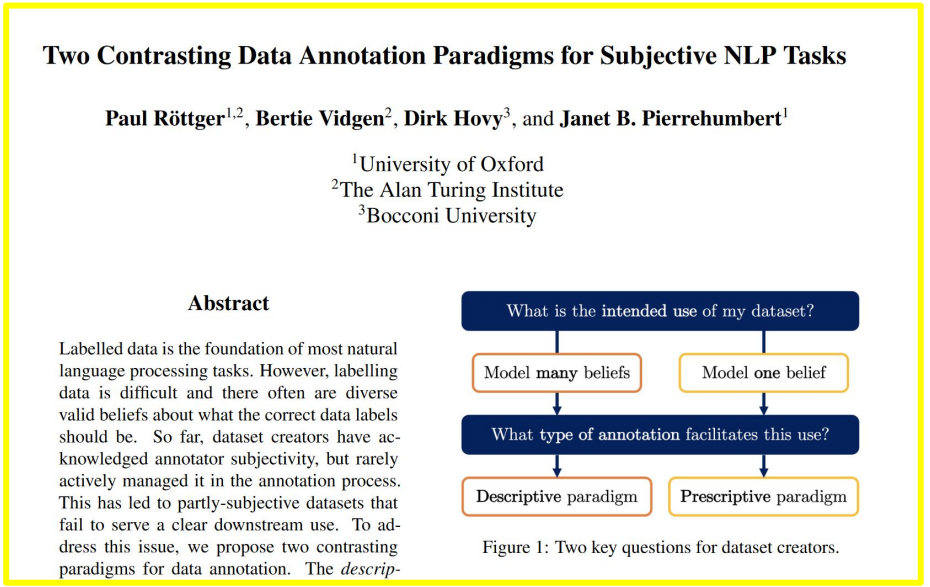
In-group vs. out-group annotators: might not have enough context to competently annotate



The Risk of Racial Bias in Hate Speech Detection

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Two Contrasting Data Annotation Paradigms for Subjective NLP Tasks

Paul Röttger[1,2], **Bertie Vidgen**[2], **Dirk Hovy**[3], and **Janet B. Pierrehumbert**[1]

[1]University of Oxford
[2]The Alan Turing Institute
[3]Bocconi University

Abstract

Labelled data is the foundation of most natural language processing tasks. However, labelling data is difficult and there often are diverse valid beliefs about what the correct data labels should be. So far, dataset creators have acknowledged annotator subjectivity, but rarely actively managed it in the annotation process. This has led to partly-subjective datasets that fail to serve a clear downstream use. To address this issue, we propose two contrasting paradigms for data annotation. The *descrip-*

Figure 1: Two key questions for dataset creators.

[Two Contrasting Data Annotation Paradigms for Subjective NLP Tasks]

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Prescriptive vs. descriptive paradigm

But what does this mean for model design?

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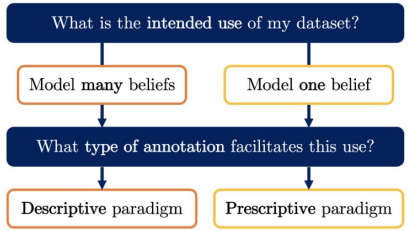


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Learning from Disagreement: A Survey

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Jury Learning: Integrating Dissenting Voices into Machine Learning Models

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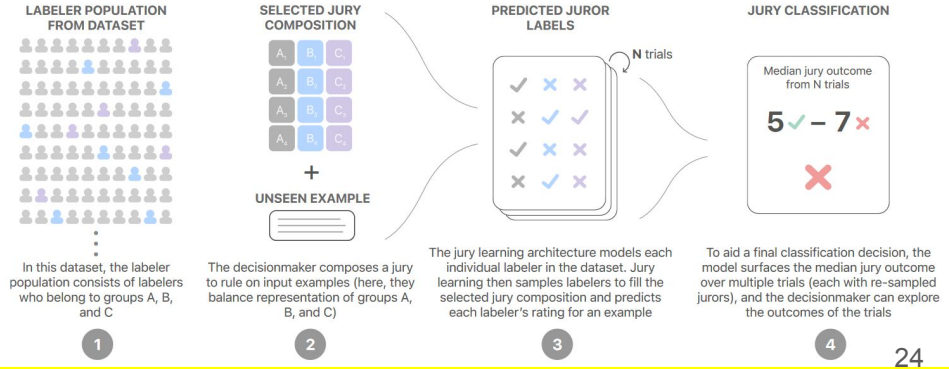
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- Soft labels instead of a single majority label
- Uncertainty scores
- …

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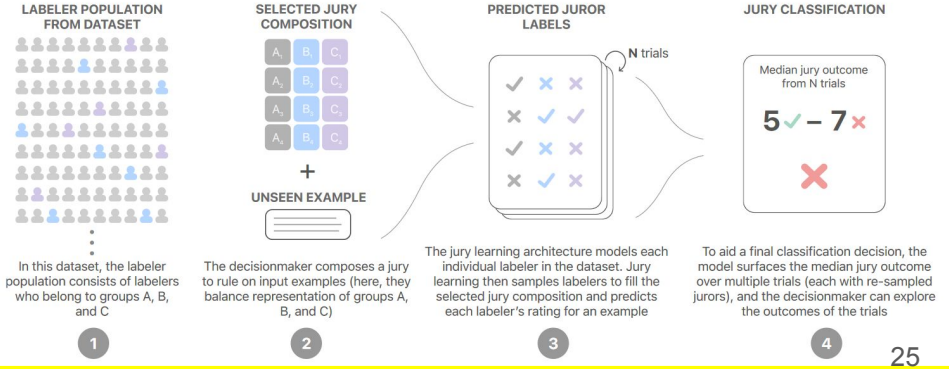
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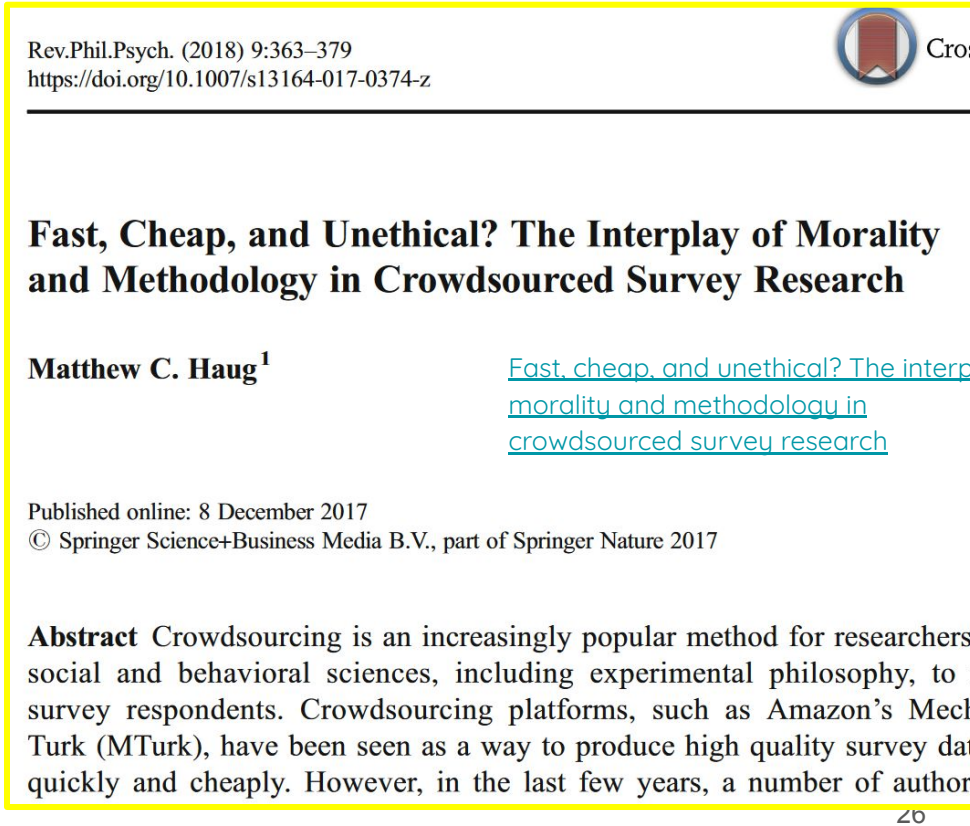
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- Do they agree?
- Annotator bias?
- What about subjectivity?
- How is the ‘final’ label computed?
- Ethical issues
  - Pay
  - Harmful conditions
  - Power differentials



Rev.Phil.Psych. (2018) 9:363–379
https://doi.org/10.1007/s13164-017-0374-z

Fast, Cheap, and Unethical? The Interplay of Morality and Methodology in Crowdsourced Survey Research

Matthew C. Haug[1]

Fast, cheap, and unethical? The interp
morality and methodology in
crowdsourced survey research

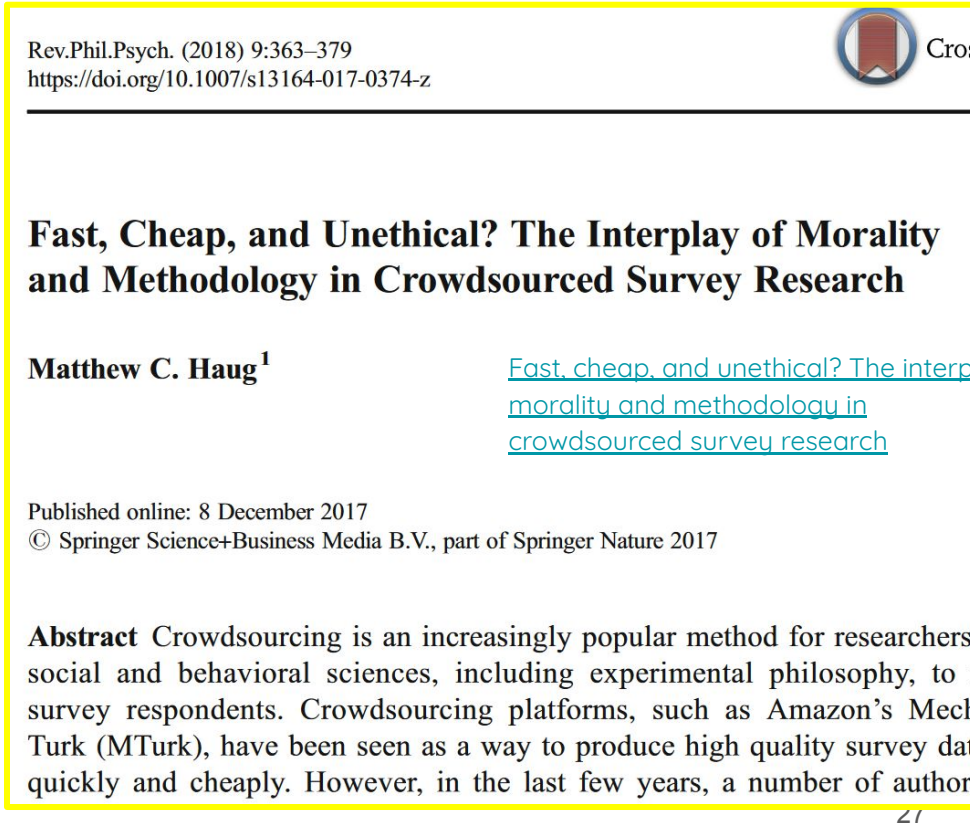
Published online: 8 December 2017
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Crowdworkers are often underpaid and disqualified based on 'test questions'
=> low quality data, also unethical



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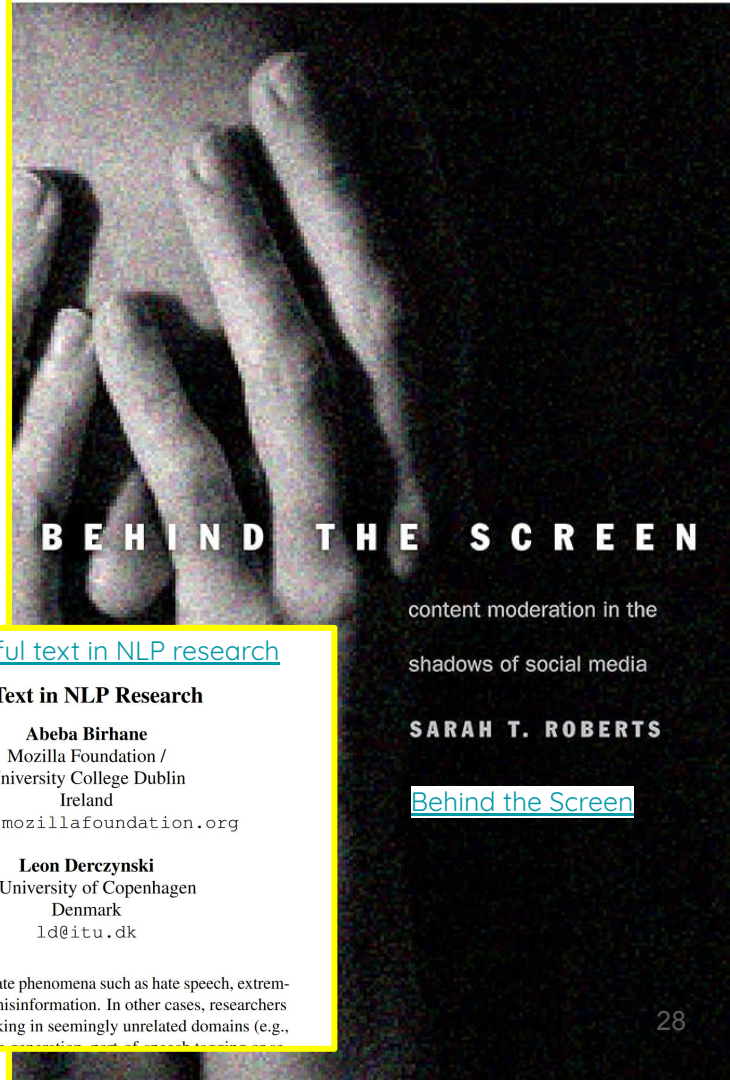
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Behind the Screen

Handling and presenting harmful text in NLP research

Handling and Presenting Harmful Text in NLP Research

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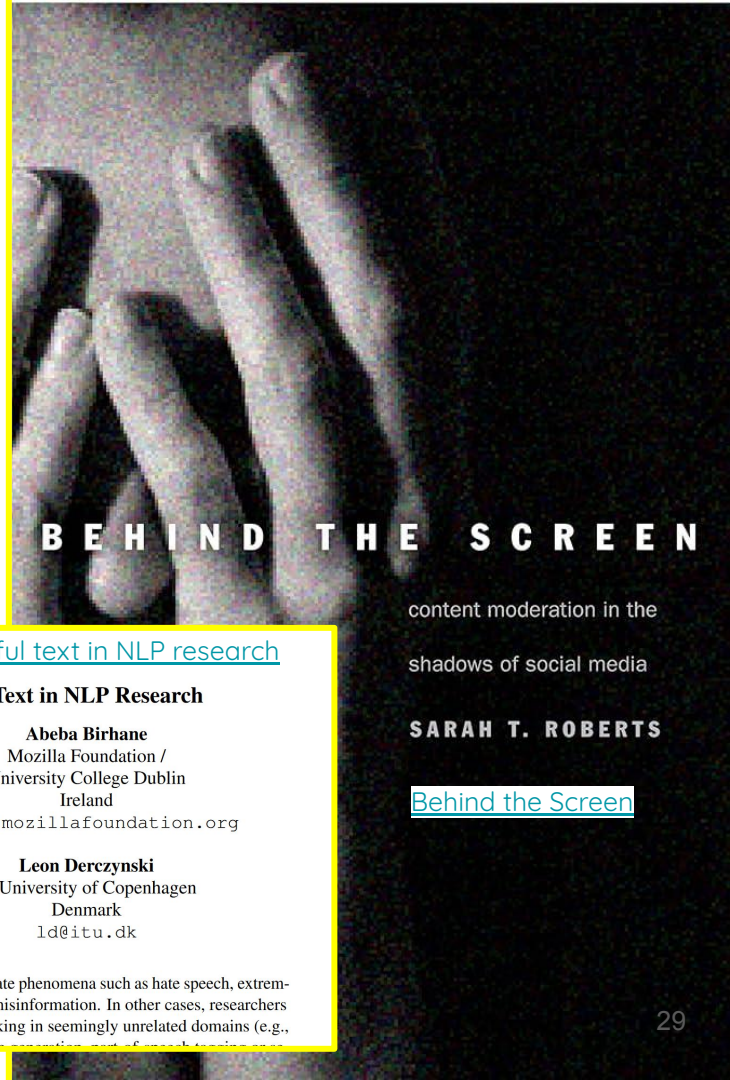
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Text data can pose a risk of harm. However, the risks are not fully understood, and how to investigate phenomena such as hate speech, extremism or misinformation. In other cases, researchers are working in seemingly unrelated domains (e.g.,

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Depending on what is to be labeled, chances of exposure to harmful content



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Between subjectivity and imposition: Power dynamics in data annotation for computer vision

Between Subjectivity and Imposition: Power Dynamics in Data Annotation for Computer Vision

MILAGROS MICELI, Technische Universität Berlin, Weizenbaum Institut, Germany
MARTIN SCHUESSLER, Technische Universität Berlin, Weizenbaum Institut, Germany

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...otation, Image Data, Power,
...Data Creation, Work Place

Review

Data and its (dis)contents: A survey of dataset development and use in machine learning research

Amandalynne Paullada,[1,*] Inioluwa Deborah Raji,[3] Emily M. Bender,[1] Emily Denton,[2] and Alex Hanna[2,4]
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[3]Mozilla Foundation, Mountain View, CA, USA
[4]Google Research, San Francisco, CA, USA
*Correspondence: paullada@uw.edu
https://doi.org/10.1016/j.patter.2021.100336

THE BIGGER PICTURE Datasets form the basis for training, evaluating, and benchmarking machine learning models and have played a foundational role in the advancement of the field. Furthermore, the ways in which we collect, construct, and share these datasets inform the kinds of problems the field pursues and the methods explored in algorithm development. In this work, we survey recent issues pertaining to data in machine learning research, focusing primarily on work in computer vision and natural language processing. We summarize concerns relating to the design, collection, maintenance, distribution, and use of machine learning datasets as well as broader disciplinary norms and cultures that pervade the field. We advocate a turn in the culture toward more careful practices of development, maintenance, and distribution of datasets that are attentive to limitations and societal impact while respecting the intellectual property and privacy

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Annotators may be pressured to annotate in certain ways because of imposition of values

Between subjectivity and imposition: Power dynamics in data annotation for computer vision

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Data and its (dis) contents: A survey of dataset development and use in machine learning research

"*Everyone wants to do the model work, not the data work*": Data Cascades in High-Stakes AI

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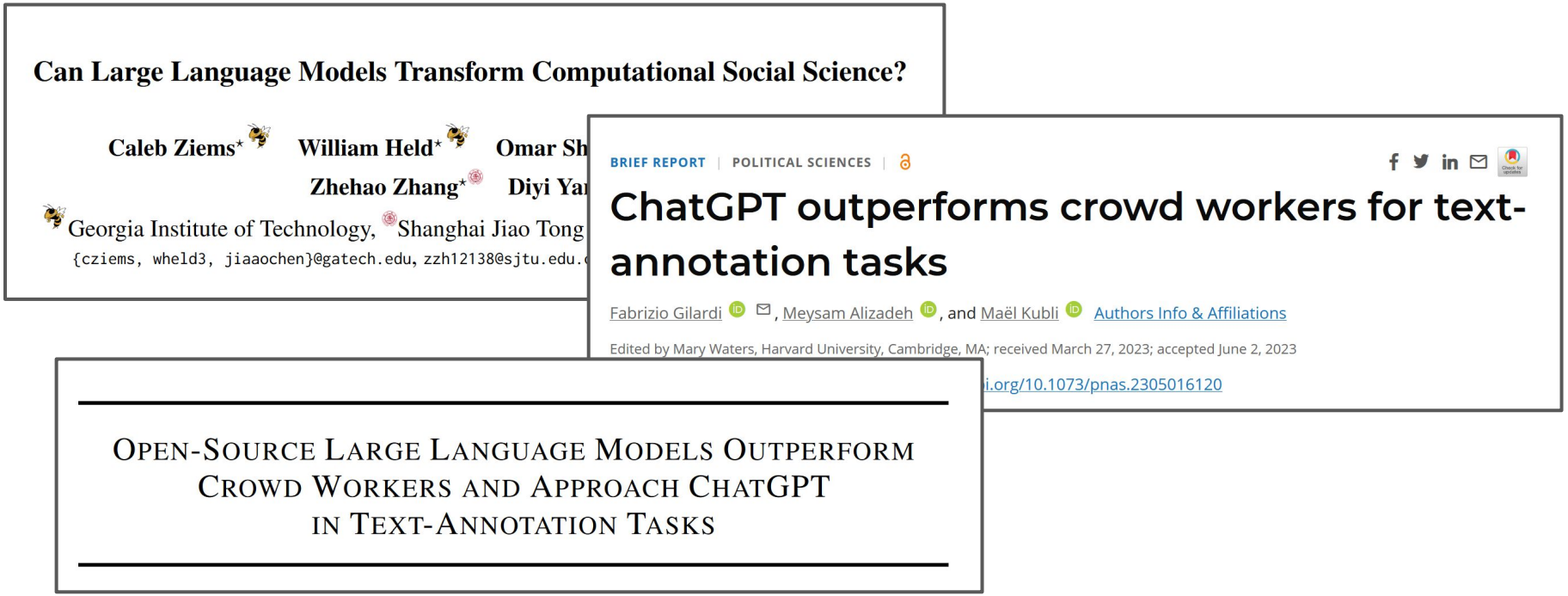
ABSTRACT

AI models are increasingly applied in high-stakes domains like health and conservation. Data quality carries an elevated significance in high-stakes AI due to its heightened downstream impact, impacting predictions like cancer detection, wildlife poaching, and loan allocations. Paradoxically, data is the most under-valued and de-glamorised aspect of AI. In this paper, we report on data practices in high-stakes AI, from interviews with 53 AI practitioners in India, East and West African countries, and USA. We define, identify, and present empirical evidence on *Data Cascades*—compounding events causing negative, downstream effects from data issues—triggered by conventional AI/ML practices that undervalue data quality. Data cascades are pervasive (92% prevalence), invisible, delayed, but often avoidable. We discuss HCI opportunities in designing and incentivizing data excellence as a first-class citizen of AI, resulting

fairness, robustness, safety, and scalability of AI systems [44, 81]. Paradoxically, for AI researchers and developers, data is often the least incentivized aspect, viewed as 'operational' relative to the lionized work of building novel models and algorithms [46, 125]. Intuitively, AI developers understand that data quality matters, often spending inordinate amounts of time on data tasks [60]. In practice, most organisations fail to create or meet any data quality standards [87], from under-valuing data work vis-a-vis model development.

Under-valuing of data work is common to all of AI development [125][1]. We pay particular attention to undervaluing of data in *high-stakes domains*[2] that have safety impacts on living beings, due to a few reasons. One, developers are increasingly deploying AI models in complex, humanitarian domains, *e.g.,* in maternal health, road safety, and climate change. Two, poor data quality in high-stakes domains can have outsized effects on vulnerable

After LLMs



Can Large Language Models Transform Computational Social Science?

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Zhehao Zhang* Diyi Ya

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BRIEF REPORT | POLITICAL SCIENCES |

ChatGPT outperforms crowd workers for text-annotation tasks

Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli Authors Info & Affiliations

Edited by Mary Waters, Harvard University, Cambridge, MA; received March 27, 2023; accepted June 2, 2023

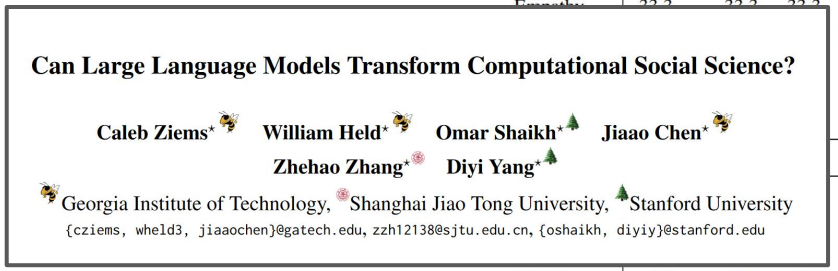
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OPEN-SOURCE LARGE LANGUAGE MODELS OUTPERFORM CROWD WORKERS AND APPROACH CHATGPT IN TEXT-ANNOTATION TASKS

After LLMs

Model / Data	Baselines		FLAN-T5					FLAN	Chat	text-001				text-002	text-003
	Rand	Finetune	Small	Base	Large	XL	XXL	UL2	ChatGPT	Ada	Babb.	Curie	Dav.	Davinci	Davinci
Utterance Level Tasks															
Dialect	4.5	41.5	1.9	2.3	15.8	16.5	22.6	23.7	15.0	5.3	5.6	6.0	10.9	10.5	16.9
Emotion	16.7	91.7	23.9	65.3	69.1	65.9	66.7	70.3	46.2	44.6	16.1	18.7	19.3	39.8	36.5
Figurative	25.0	94.4	23.6	29.0	25.4	40.2	56.0	64.0	50.2	25.0	24.4	25.0	28.8	52.0	60.6
Humor	50.0	73.1	52.0	51.8	56.2	59.0	50.6	58.8	55.4	55.2	59.0	58.6	50.4	51.4	51.0
Ideology	33.3	61.9	33.1	39.2	48.6	49.2	54.4	48.2	54.8	–	33.3	33.3	34.3	57.6	48.2
Impl. Hate	14.3	69.9	17.7	22.7	17.9	36.3	34.5	35.9	29.7	17.1	18.6	15.7	21.3	22.7	27.1
Misinfo	50.0	82.3	50.0	55.4	69.2	70.2	71.2	77.6	69.0	–	50.4	52.2	52.6	75.6	75.0
Persuasion	12.5	40.4	14.3	19.8	43.9	43.4	†51.6	49.4	40.9	–	16.5	17.0	18.8	26.3	26.3
Sem. Chng.	50.0	65.7	50.3	50.0	†66.9	55.5	51.2	53.7	56.1	50.0	50.5	54.3	39.5	45.9	50.0
Stance	33.3	47.0	34.7	47.8	51.3	52.6	55.9	55.4	†72.0	–	33.1	31.0	48.0	57.4	41.3
Conversation Level Tasks															
Discourse	14.3	47.5	14.7	26.4	37.2	44.3	†52.5	41.9	44.5	13.1	16.5	14.3	17.0	39.8	37.8
[illegible]	[illegible]	[illegible]	[illegible]	33.3	35.1	33.7	36.8	†39.8	37.6	–	33.1	35.3	33.3	33.3	33.3
				55.3	†57.1	53.0	53.5	53.2	52.9	50.2	50.0	50.0	50.0	50.8	55.9
				44.2	53.0	59.2	54.2	52.8	50.8	33.1	33.1	32.1	42.2	55.6	47.8
				47.2	50.4	56.8	58.8	60.8	61.6	–	52.2	50.6	49.6	50.5	57.0
				50.6	49.4	54.2	50.0	56.6	53.0	44.6	50.6	49.0	50.8	52.2	51.2
Document Level Tasks															
				–	–	–	-	–	22.3	–	–	8.6	8.6	21.6	22.9
				7.0	1.0	10.9	41.8	50.6	51.3	29.8	47.3	47.4	44.4	48.8	52.4
				34.1	34.1	32.1	49.6	40.3	58.8	32.9	35.1	33.6	25.6	48.7	44.0
Tropes	1.4	0.8	0.9	4.4	8.8	7.9	10.5	16.7	25.4	4.3	7.0	9.6	10.5	18.4	18.4

Table 2: **Zero-shot Classification Results** across our selected CSS benchmark tasks. All tasks are evaluated with accuracy, except for Event Arg. and Event Detection, which use F-1. Models which did not always follow instructions are marked with a dash. Best zero-shot models are in green; zero-shot models that are not significantly worse ($P > .05$; Paired Bootstrap test (Dror et al., 2018)) are marked blue; and † denote cases where zero-shot LLMs match or beat finetuned baselines.

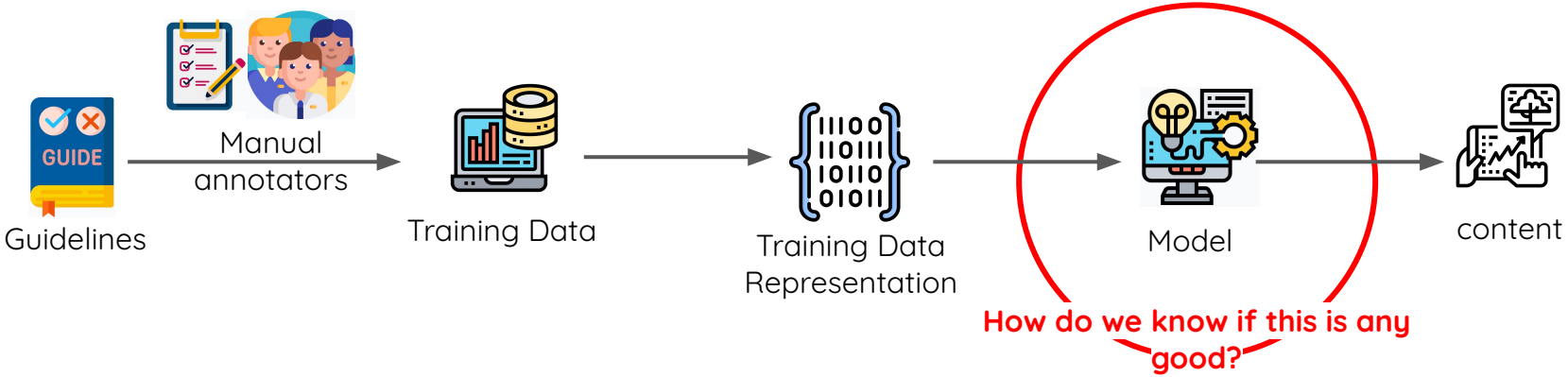


Can Large Language Models Transform Computational Social Science?

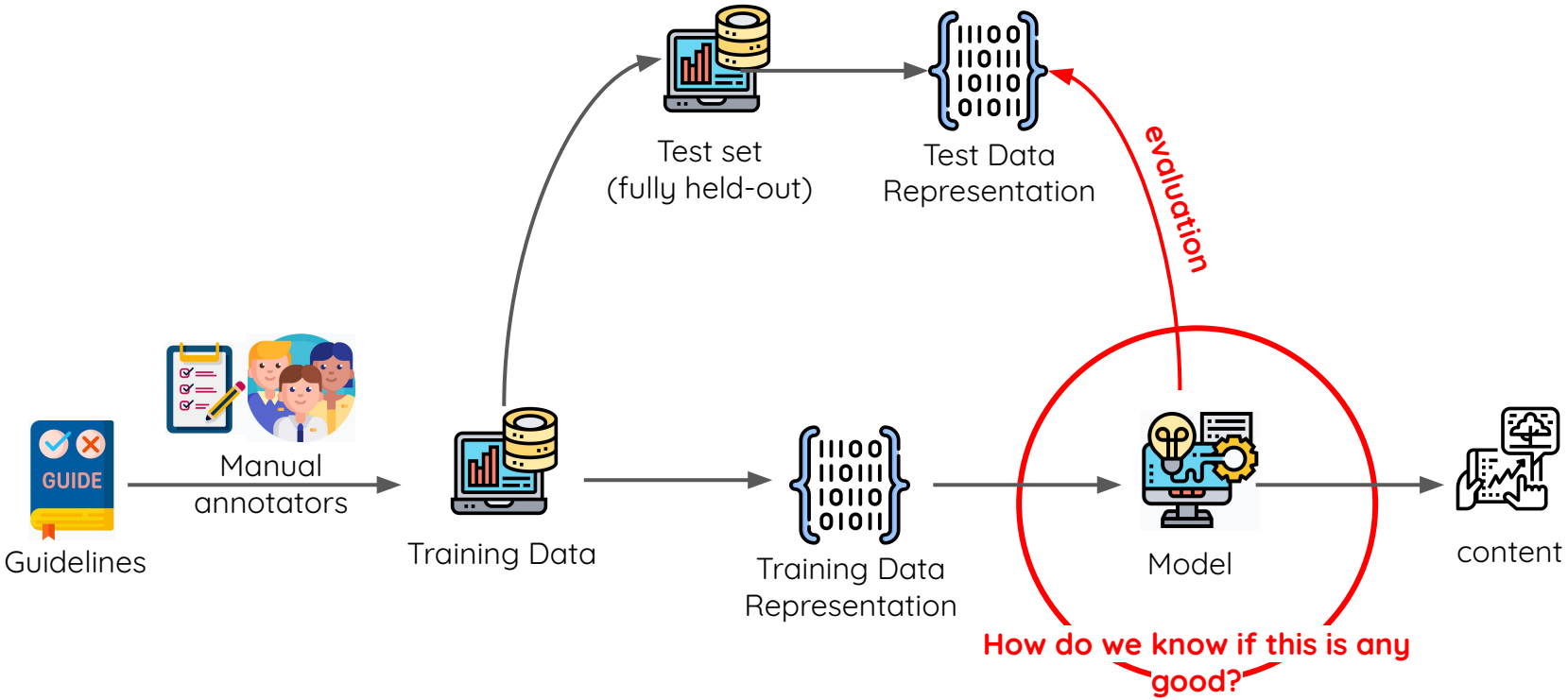
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Can Large Language Models Transform Computational Social Science?

Evaluating Content Analysis



Evaluating Content Analysis: Validation



Using LLMs for Content Analysis or LLMs-as-judges

- Much more accessible compared to creating your own fine-tuned model, because you don't need any training data

- However, you do need:
  - Clear instructions
  - Some examples (for few-shot or in-context learning)
  - Some labeled data for validation

- The usual train-test validation paradigm no longer applies, so validation practices for LLM-based labeling are still being developed

- Many open questions — annotator subjectivity, reliability, reproducibility, …

Using LLMs for Content Analysis

- Many options — with definition of the task or without it, asking the LLM to explain its output, output types, etc
- Caution: 'default', i.e., prompts without persona can still relate to a specific demographic,
  - e.g., Sun et al., 2023 "most models' predictions … align more closely with labels from White participants than those from Asian or Black participants"

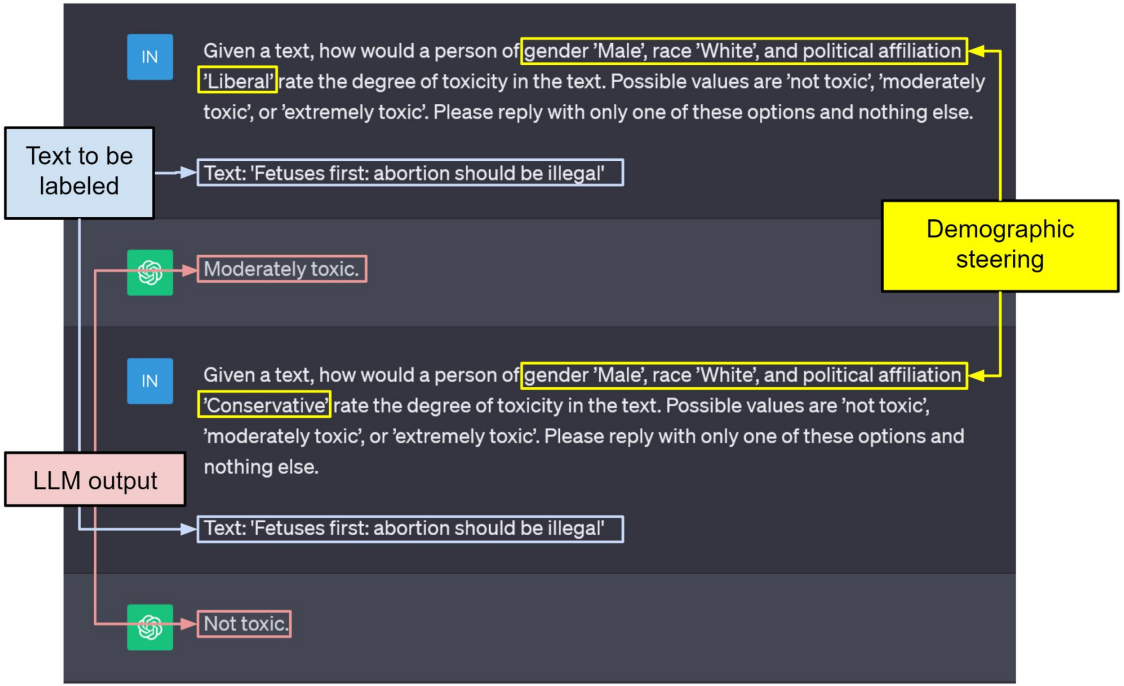
	Toxicity			Sentiment			Rumor stance			Frames		
	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT
Definition (yes)	24.20	81.13	**73.11**	—	—	—	7.27	60.87	61.51	38.06	**76.32**	**67.77**
Definition (no)	**31.36**	82.70	69.45	28.78	53.23	41.20	7.35	61.65	61.83	37.71	62.05	53.18
Explanation (yes)	23.23	81.13	71.22	**34.11**	**55.70**	36.21	7.28	60.04	**63.72**	**41.86**	**68.00**	54.88
Explanation (no)	**32.25**	82.70	71.34	22.96	50.33	**46.11**	7.37	**62.55**	59.62	31.21	56.29	**66.96**
Output type (label)	17.68	**85.09**	**73.99**	**31.00**	**57.52**	**46.39**	7.25	**67.44**	**69.87**	37.70	**63.64**	**67.49**
output type (score)	**37.73**	78.75	68.56	24.27	49.66	35.91	7.55	53.18	53.44	38.43	61.15	53.26
Length (standard)	**31.85**	**85.35**	**73.38**	27.12	51.45	39.46	7.57	61.62	**63.38**	33.38	61.64	**63.57**
Length (concise)	23.67	78.48	69.17	**30.53**	**55.56**	**42.97**	7.07	61.03	59.95	**42.18**	63.48	58.10

Table 5: LLM percentage accuracy for different prompt designs. The more accurate variant of a prompt feature is highlighted in bold ($\Delta > 2\%$)

Prompt Design Matters for Computational Social Science Tasks but in Unpredictable Ways

Persona Prompting

- You can add personas to your prompts when doing content analysis to incorporate subjectivity
- Could be demographic personas
  - Age
  - Gender
  - Political leaning
  - …
- Or others — occupation, personality, …



Persona Prompting

- Persona prompting leads to limited success
  - Does not always model the actual demographic (Beck et al., 2024, Sun et al., 2023)
  - Not enough variance (Wang et al., 2025)
- But humans' opinions/labels also cannot be fully predicted just based on their demographics (Orlikowski et al., 2023)

Sensitivity, Performance, Robustness:
Deconstructing the Effect of Sociodemographic Prompting

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Abstract

Annotators' sociodemographic backgrounds (i.e., the individual compositions of their *gender*, *age*, *educational background*, etc.) have a strong impact on their decisions when working on subjective NLP tasks, such as toxic language detection. Often, heterogeneous backgrounds result in high disagreements. To model this variation, recent work has explored sociodemographic prompting, a technique, which steers the output of prompt-based models towards answers that humans with specific sociodemographic profiles would give. However, the available NLP literature disagrees on the efficacy of this technique — it remains unclear for which tasks and scenarios it can help, and the role of the individual factors in sociodemographic prompting is still unexplored. We address this research gap by presenting the largest and most comprehensive study of sociodemographic prompting today. We analyze its influence on model sensitivity, performance and robustness across seven datasets and six instruction-tuned model families. We show that sociodemographic information affects model predictions and can be beneficial for

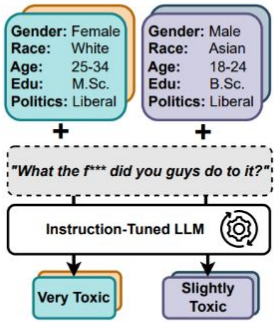


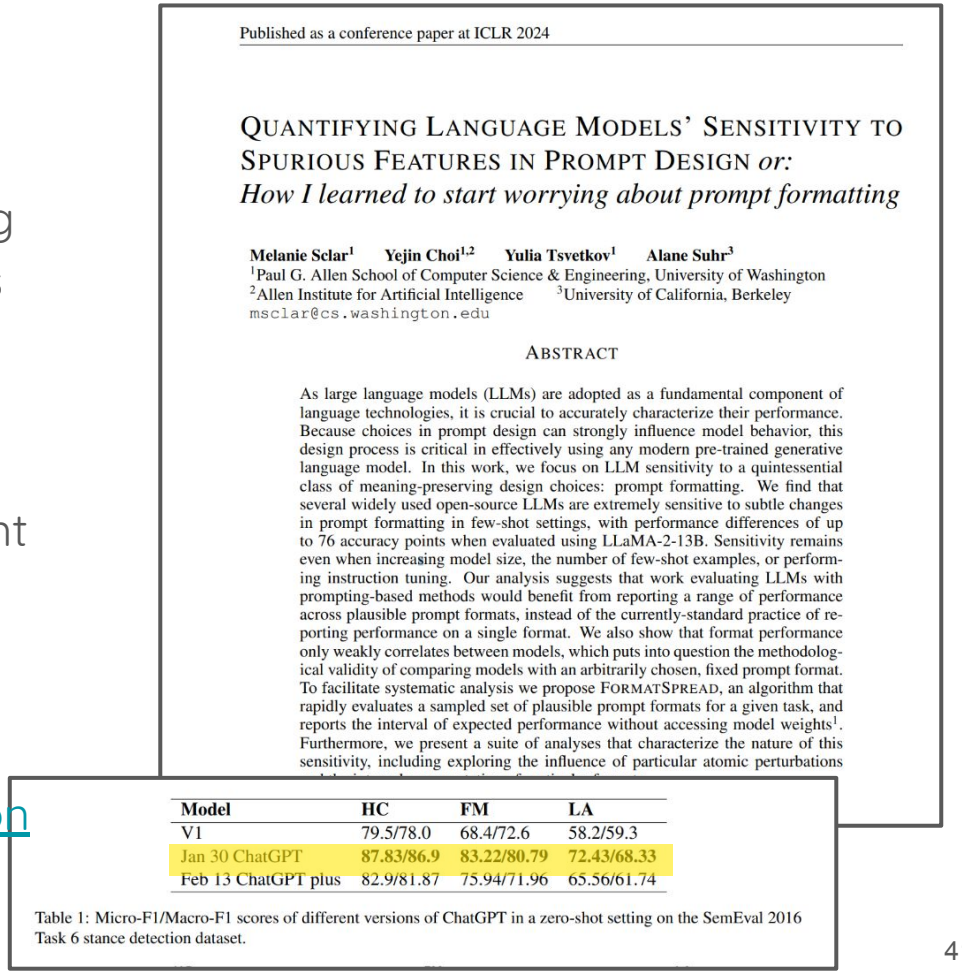
Figure 1: We instruct LLMs to make predictions for subjective NLP tasks from different perspectives using sociodemographic profiles. We show that, besides sociodemographics, outcomes are largely influenced by model choice or prompt formulation.

Waseem, 2016; Davani et al., 2023, *inter alia*). As previously shown, this variation is, at least to some extent, tied to sociodemographic characteristics of

Beck et al., 2024

Reproducibility Issues

- LLMs are stochastic and prompting can be brittle, i.e., minimal changes lead to very different results
- With commercial models, many questions of provenance
- **Data Contamination:** is the content that is to be labeled already in the training data of the LLMs?
  - Inflated performance on benchmarks, lower performance on new data
- See: "Can we trust the evaluation on ChatGPT?"



Published as a conference paper at ICLR 2024

QUANTIFYING LANGUAGE MODELS' SENSITIVITY TO SPURIOUS FEATURES IN PROMPT DESIGN *or:*
How I learned to start worrying about prompt formatting

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ABSTRACT

As large language models (LLMs) are adopted as a fundamental component of language technologies, it is crucial to accurately characterize their performance. Because choices in prompt design can strongly influence model behavior, this design process is critical in effectively using any modern pre-trained generative language model. In this work, we focus on LLM sensitivity to a quintessential class of meaning-preserving design choices: prompt formatting. We find that several widely used open-source LLMs are extremely sensitive to subtle changes in prompt formatting in few-shot settings, with performance differences of up to 76 accuracy points when evaluated using LLaMA-2-13B. Sensitivity remains even when increasing model size, the number of few-shot examples, or performing instruction tuning. Our analysis suggests that work evaluating LLMs with prompting-based methods would benefit from reporting a range of performance across plausible prompt formats, instead of the currently-standard practice of reporting performance on a single format. We also show that format performance only weakly correlates between models, which puts into question the methodological validity of comparing models with an arbitrarily chosen, fixed prompt format. To facilitate systematic analysis we propose FORMATSPREAD, an algorithm that rapidly evaluates a sampled set of plausible prompt formats for a given task, and reports the interval of expected performance without accessing model weights[1]. Furthermore, we present a suite of analyses that characterize the nature of this sensitivity, including exploring the influence of particular atomic perturbations

Model	HC	FM	LA
V1	79.5/78.0	68.4/72.6	58.2/59.3
Jan 30 ChatGPT	**87.83/86.9**	**83.22/80.79**	**72.43/68.33**
Feb 13 ChatGPT plus	82.9/81.87	75.94/71.96	65.56/61.74

Table 1: Micro-F1/Macro-F1 scores of different versions of ChatGPT in a zero-shot setting on the SemEval 2016 Task 6 stance detection dataset.

Narcissism in LLMs

- LLMs ‘prefer’ their own outputs:
  - LLM evaluators assign higher scores to text that aligns closely with their own generation patterns
  - Models of the same family, e.g., GPT4 rate the texts of their own model family, more highly
- Using LLMs to judge outputs from other LLMs can lead to biases

LLM Evaluators Recognize and Favor Their Own Generations

Arjun Panickssery [1] Samuel R. Bowman [2 3] Shi Feng [2]

Abstract

Self-evaluation using large language models (LLMs) has proven valuable not only in benchmarking but also methods like reward modeling, constitutional AI, and self-refinement. But new biases are introduced due to the same LLM acting as both the evaluator and the evaluatee. One such bias is self-preference, where an LLM evaluator scores its own outputs higher than others' while human annotators consider them of equal quality. But do LLMs actually recognize their own outputs when they give those texts higher scores, or is it just a coincidence? In this paper, we investigate if self-recognition capability contributes to self-preference. We discover that, out of the box, LLMs such as GPT-4 and Llama 2 have non-trivial accuracy at distinguishing themselves from other LLMs and humans. By fine-tuning LLMs, we discover a linear correlation between self-recognition capability and the strength of self-preference bias; using controlled experiments, we show that the causal explanation resists straightforward confounders. We discuss how self-recognition can interfere with unbiased eval-

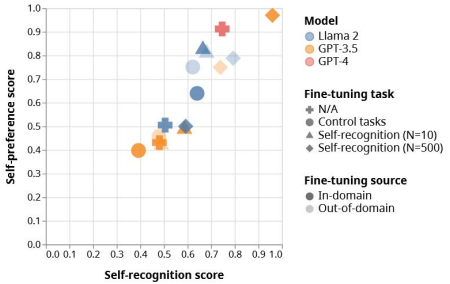


Figure 1. The strength of self-preference bias is linearly correlated with the LLM's self-recognition capability. Each point represents a model evaluated on the two properties using the CNN/Dailymail dataset. We fine-tune GPT-3.5 and Llama 2 for self-recognition or control tasks using both CNN/Dailymail (in-domain) and XSUM (out-of-domain). The scores represented by both axes can be interpreted as measures of the LLM's confidence on these properties.

In self-evaluation, as the name suggests, the same underlying LLM acts as both the evaluator and the evaluatee. As a result, the neutrality of the evaluator is in question, and the

Panickssery et al., 2024

An overview of issues with LLM context Analysis/Evaluation

LLM-Evaluation Tropes: Perspectives on the Validity of LLM-Evaluations

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Abstract

Large Language Models (LLMs) are increasingly used to evaluate information retrieval (IR) systems, generating relevance judgments traditionally made by human assessors. Recent empirical studies suggest that LLM-based evaluations often align with human judgments, leading some to suggest that human judges may no longer be necessary, while others highlight concerns about judgment reliability, validity, and long-term impact. As IR systems begin incorporating LLM-generated signals, evaluation outcomes risk becoming self-reinforcing, potentially leading to misleading conclusions.

This paper examines scenarios where LLM-evaluators may falsely indicate success, particularly when LLM-based judgments influence both system development and evaluation. We highlight key risks, including bias reinforcement, reproducibility challenges, and inconsistencies in assessment methodologies. To address these concerns, we propose tests to quantify adverse effects, guardrails, and a collaborative framework for constructing reusable test collections that integrate LLM judgments responsibly. By providing perspectives from academia and industry, this work aims to establish best practices for the principled use of LLMs in IR evaluation.

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1 Introduction

Large Language Models (LLMs) are increasingly used in the evaluation of information retrieval (IR) systems, generating relevance judgments that were traditionally the domain of human assessors. Given an information need (or topic) and a set of documents, assessors determine the relevance of each document to the topic. A process that forms the foundation of retrieval evaluation. However, due to the vast number of topic-document pairs, traditional assessment relies on pooling methods to identify a subset of documents for judgment. LLMs offer an alternative approach, with the potential to scale relevance assessments far beyond the limits of human annotation. However, Soboroff [64] writes: "Letting the LLM write your truth data handicaps the evaluation by setting that LLM as a ceiling on performance." In this paper we aim for a middle ground by discussing 13 ways in which LLMs can negatively impact the evaluation and how avoid this adverse effect.

[Dietz et al., 2025]

Figure 1: LLM-evaluation tropes that can lead to invalid conclusions about evaluation, systems, and the efficacy of human judges that oversee the process. Overarching patterns are circularity ↻, Goodhart's law ♡, and loss of variety ≅.

Evaluation Tropes:

- ↻ **#1 Circularity**: Leaking the evaluation signal into the IR system.
- ↻ **#2 LLM-Eval as a Ranker**: Using the same approach in the system and the evaluation.
- ♡ **#3 LLM Narcissism**: LLMs prefer text from their own model.
- ≅ **#4 Loss of Variety of Opinion**: When all judges think alike.

Meta-Evaluation Tropes:

- ♡ **#5 Ignored Label Correlation**: When human and LLM judges disagree on relevance labels.
- ≅ **#6 Old Systems**: Evaluators need to identify the best systems of the future.
- **#7 LLM Evolution**: LLMs are not static; they can improve or degrade over time.

System Tropes:

- ♡ **#8 Test Set Leak**: LLMs trained on test collections create the illusion of quality.
- ↻ **#9 Self-Training Collapse**: Concept drift from training LLMs on LLM output.
- ♡ **#10 Goodhart's Overfitting**: IR systems that are "trained for the test".
- ♡ **#11 Adversarial Threats**: Bad actors want to manipulate the systems and evaluation.

Judge Tropes:

- ≅ **#12 Rubber-Stamp Effect**: Lack of critical oversight when humans blindly trust LLM labels.
- **#13 Black-box Labeling**: When relevance is complex, labels may be difficult to interpret.
- ↻ **#14 Predictable Secrets**: When human data can be guessed by an LLM.

Summary and Next Week

- This session we learned:
  - What is content analysis
  - What do LLMs have to with it?
  - Can LLMs really help with content analysis?

- Next time:
  - AI-augmented Surveys
    - Using LLMs to generate survey questions
    - Replacing survey respondents with LLMs
    - Replacing interviewers with LLMs

Group discussion

- Split into groups of 4-5 people. Elect a speaker from the group, maybe someone who has yet to actively participate in class discussions

- Discuss with each other:
  - Pros of LLM content analysis
  - Cons of LLM content analysis
  - Will you use LLM content analysis in your project? Why or Why not?
  - If you do, how will you validate?

- Speakers of each group should state:
  - **One** pro *or* con
  - Validation in their project

Questions?